



Aerodynamic shape optimization with effective deep-learning-based geometric filtering: from a cylinder to a wing

Yan Chen^{1,2} · Jichao Li^{1,2} · Chao Song³ · Hongyang Liu³ · Zhu Zhou³ · Jiangtao Huang⁴ · Jinsheng Cai^{1,2} · Boo Cheong Khoo⁵

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Abstract

Geometric filtering with deep-learning-based validity constraints has proven useful in improving optimization convergence efficiency via shrinking geometric space. Nevertheless, in challenging aerodynamic shape design scenarios, for example, reducing the drag of bluff shapes, it is ineffective to directly embed geometric validity constraints into optimizers, and the optimization tends to fail even after a long iteration struggle. To address the issue, we propose a hybrid gradient-based aerodynamic shape optimization method that effectively leverages deep-learning-based geometric filtering to enhance optimization convergence. In this method, the optimization problem is first approximately solved using aerodynamic sensitivities and geometric validity constraints. To enhance filtering effectiveness, we improve the validity model generalizability by augmenting airfoil shape datasets. After the optimization reaches an optimal geometric sub-domain, an improved gradient-based optimization is switched on to accurately solve the problem. The effectiveness of this approach is validated in aerodynamic shape optimization of two-dimensional and three-dimensional cylindrical shapes. Both optimization cases converge smoothly, highlighting much higher effectiveness compared with the state-of-the-art aerodynamic shape optimization framework. This work facilitates effective aerodynamic shape optimization in extensively large geometric design spaces, establishing a foundation to achieve more efficient aerodynamic designs.

Keywords Aerodynamic optimization · Artificial intelligence · Airfoil design · Wing design

1 Introduction

Aerodynamic shape optimization (ASO) is an effective design approach to improving aerodynamic efficiency (Jameson 1988; Martins 2022), along with the advances of Computational Fluid Dynamics (CFD). Various ASO methods,

such as adjoint-based optimization (Jameson 1988; Reuther et al. 1999; Lyu et al. 2015) and surrogate-based optimization (Queipo et al. 2005; Han et al. 2019; Koziel and Leifsson 2013; Li and Zhang 2021a), have been developed to meet diverse design requirements, and these methods have shown to be effective in design optimization of airfoils (He et al. 2019; Lyu et al. 2024), wings (Bartoli et al. 2019; Bons et al. 2019; Yu et al. 2018; Jacome and Elham 2017) and aircraft (Kenway and Martins 2014; Chai et al. 2018; Liem et al. 2015). Recent development in deep learning (LeCun et al. 2015; Zhang et al. 2018; Pang and Cao 2024) further enhances ASO in terms of design efficiency (Li et al. 2019; Li and Zhang 2021b) and robustness (Li et al. 2019, 2022a, b). Nevertheless, most of these methods still put strict restrictions on a modest geometric design space, otherwise, the optimization may easily fail. These restrictions affect both optimization robustness and practical applicability. An optimization approach that can effectively search in high-dimensional geometric design space and robustly converge even starting from arbitrary bluff shapes is of great interest.

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✉ Jichao Li
jjchaoli@nwpu.edu.cn

¹ School of Aeronautics, Northwestern Polytechnical University, Xi'an 710072, China

² National Key Laboratory of Aircraft Configuration Design, Xi'an 710072, China

³ Computational Aerodynamics Institute, China Aerodynamics Research and Development Center, Mianyang 621000, China

⁴ Aerospace Technology Institute, China Aerodynamic Research and Development Center, Mianyang 621000, China

⁵ National University of Singapore, Singapore 117575, Singapore

ASO using the adjoint method (Jameson 1988; Mader et al. 2008; Kenway et al. 2019) and a gradient-based optimization algorithm (Nocedal and Wright 2006) is efficient in solving high-dimensional problems because the computational cost scales linearly with the number of design variables (Lyu et al. 2014). This facility makes it widely used in ASO of complex three-dimensional shapes such as aircraft (Kenway and Martins 2016), wind turbine blades (Madsen et al. 2019), and vehicles (He et al. 2018). Optimization with hundreds and even thousands of geometric design variables can be effectively solved. Nevertheless, adjoint-based optimization in large geometric design space still lacks robustness. He et al. (2019) performed adjoint-based optimization of a two-dimensional (2D) circle to reduce the drag in a transonic regime and encountered convergence failures even using the state-of-the-art ASO framework. The main issue stemmed from the occurrence of abnormal shapes when searching the high-dimensional, large design space, such as wavy airfoil surfaces. This is referred to as the geometric abnormality issue in the literature (Li et al. 2022a). Geometric abnormalities led to failures of CFD mesh deformation and CFD simulation convergence, which finally made the optimization fail. He et al. (2019) managed to avoid these geometric issues by developing an adaptive free-form deformation (FFD) method, which weakened the problem difficulty by reducing the number of design variables to a low-dimensional level. Wei et al. (2024) addressed this problem by proposing shape parameterization using a deep neural network (DNN) with geometric regularization, called DeepGeo. DNN parameters were treated as geometric design variables, which altered the aerodynamic shape in a global way. Geometric abnormality issues can be avoided using DeepGeo. Nevertheless, the number of DNN parameters can be millions, and this approach led to a super high-dimensionality optimization problem that can be hardly solved using classic gradient-based optimization algorithms such as Quasi-Newton methods. Wei et al. (2024) adopted a stochastic gradient descent method to approximately solve the 2D circle optimization problem and managed to obtain an effective design after eliminating aerodynamic constraints. He et al. (2019) and Wei et al. (2024) sought to address this challenging problem by advancing geometric parameterization from different perspectives, i.e., adaptively adding geometric freedoms and leveraging geometric regularization, respectively.

Surrogate-based optimization cannot scale well with the increase of dimensionality and bounds due to the *curse of dimensionality* issue, which significantly restricts the scope of surrogate-based optimization. Deep learning provides significantly effective approaches to efficient shape parameterization via extracting low-dimensional latent space, which advances surrogate-based optimization to challenging ASO problems (Li et al. 2022a, 2024; Li and Zhang

2021a). The fundamental assumption of these deep-learning-based parameterization methods is that optimal aerodynamic shapes lie in a low-dimensional manifold (Li et al. 2022a), although hundreds of conventional geometric design variables are required to find them (Lyu et al. 2015). Most of these methods are validated in airfoil shape design optimization (Chen et al. 2019; Du et al. 2021; Wang et al. 2023) because of the publicly accessible airfoil coordinates database contributed by the University of Illinois at Urbana–Champaign (UIUC). For three-dimensional (3D) aerodynamic shapes, there are rarely such adequate databases either in academia or industry, and the bottleneck of using deep-learning-based parameterization is to establish an informative database (Li et al. 2022a; Duan et al. 2019; Chen and Ramamurthy 2021). For example, Li et al. (2020) developed a geometric abnormality analysis model leveraging deep learning, which can quickly and accurately detect various geometric abnormalities, and using the abnormality model, Li and Zhang (2021a) further proposed a constrained optimal sampling method to quickly create a large database of reasonable aerodynamic shapes by filling the feasible geometric design space. This approach enables effective low-dimensional parameterization and efficient surrogate-based optimization of 3D shapes such as aircraft wings (Li and Zhang 2021a) and wind turbine blades (Li et al. 2024). Nevertheless, the methodology still exhibits a lack of effectiveness in challenging scenarios such as optimization of a bluff shape that requires effective searching of a high-dimensional design space with large bounds.

Geometric abnormality is one of the most significant issues restricting effective optimization. Deep-learning-based aerodynamic shape validity analysis provides a way for geometric filtering, preventing optimization from searching and evaluating abnormal aerodynamic shapes that are inefficient and useless. With this philosophy, Li and Zhang (2021c) applied the deep-learning-based geometric validity model to adjoint-based optimization of a 2D circle, and the drag reduction optimization converged smoothly. Nevertheless, this direct coupling approach is not always effective. Failures were encountered when solving more complicated problems. The reasons are twofold. On the one hand, the deep-learning-based validity model still lacks generalizability and thus is not always reliable. On the other hand, geometric validity cannot be ensured in line searching processes of adjoint-based optimization, and thus, there are still abnormal shapes fed to CFD evaluations, which trigger many evaluation failures. This issue becomes significant in high-dimensional 3D shape optimization and weakens the effectiveness of adjoint-based optimization.

To improve optimization robustness in challenging scenarios, we propose a hybrid gradient-based aerodynamic shape optimization approach with effective deep-learning-based geometric filtering. To enhance the generalizability of the

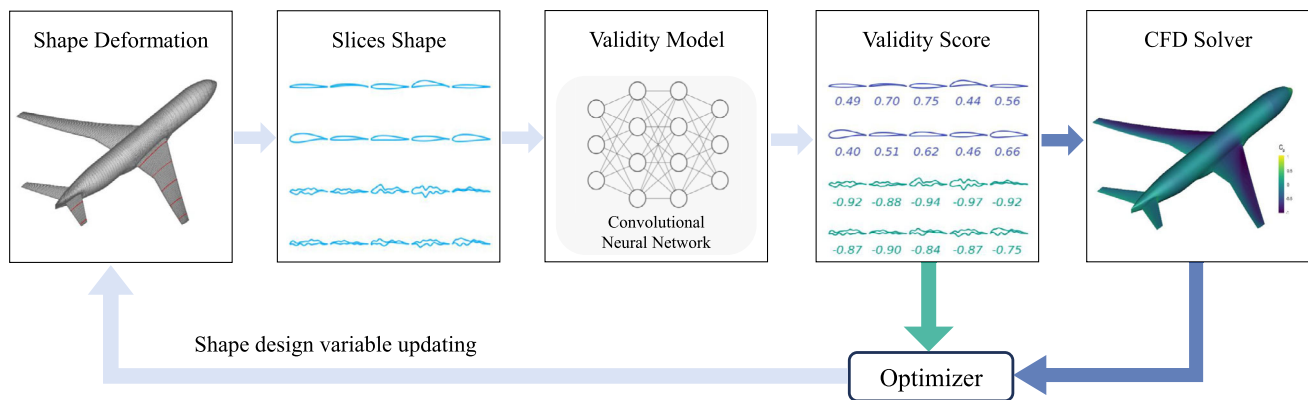


Fig. 1 Deep-learning-based geometric filtering enables efficient aerodynamic shape optimization

deep-learning-based geometric validity model, we refine the training databases by advancing the airfoil generative model and improving shape deformation settings. Then, instead of directly coupling geometric filtering with adjoint-based optimization, we first approximate the optimization problem to a series of sub-optimization problems using a trust region strategy. In each sub-optimization iteration, aerodynamic performances are approximated using the adjoint method and the first-order Taylor expansion. The sub-optimization problem is solved inside the trust region subject to geometric validity and aerodynamic constraints. After reaching a convergence criterion, an improved gradient-based optimization with strict constraints on geometric validity and aerodynamic performance is triggered. This hybrid method bypasses the difficulty in large-scale geometric design space via effective deep-learning geometric filtering and provides effective solutions to ASO in this scenario.

The rest of the paper is organized as follows. First, we introduce how we improve the generalizability of the deep-learning-based geometric filtering model in Sect. 2. Then, we present the hybrid optimization framework in Sect. 3. We show the optimization effectiveness in 2D circle and 3D cylinder cases in Sects. 4 and 5, respectively. Finally, in Sect. 6, we give a summary of conclusions.

2 Deep-learning-based geometric validity analysis model

Geometric validity models provide fast evaluations of aerodynamic shapes without the need to call for CFD simulations. Deep learning is an effective way to build such a model via a data-driven approach. Figure 1 briefly shows the philosophy of using a deep-learning-based shape validity model in aerodynamic shape optimization. The model filters out abnormal shapes before CFD evaluations and thus improves optimization efficiency and effectiveness.

Li et al. (2020) presented an airfoil validity model to identify abnormal sectional shapes in the design optimization of aircraft wings and wind turbine blades. The model was trained with realistic airfoils and abnormal shapes, and its effectiveness has been validated in aerodynamic shape optimization of airfoils (Li et al. 2020, 2022), wings (Li and Zhang 2021a, c), and wind turbine blades (Li et al. 2024). Nevertheless, the model developed in previous work (Li et al. 2020) has a lack of generalizability, and this issue leads to ineffective geometric filtering and even optimization failures. In this section, we enhance the model's generalizability by refining training databases.

2.1 Airfoil shape database construction using a generative model

Open accessible airfoil shape data are limited. We use Generative Adversarial Networks (GAN) to augment the available airfoil database. As shown in Fig. 2, GAN is composed of a generator G and a discriminator D . The former is designed to learn the distribution of underlying latent variables from the training data and thus generates similar synthetic data. The latter is trained to determine whether a given sample is real or synthetic. G and D are trained adversarially and simultaneously, with a two-player minimax function V measuring Nash equilibrium (Goodfellow et al. 2014). That is,

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim P_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim P_z(z)} [\log(1 - D(G(z)))]. \quad (1)$$

The generator of GAN can be used to enhance the database for the training of data-driven geometric validity models. Nevertheless, GAN may suffer from the mode collapse issue that makes synthetic data lose diversity, when the generator merely learns a part of the data distribution. Wasserstein GAN (WGAN; Arjovsky et al. 2017) uses the Wasserstein distance to measure data differences, and it provides mean-

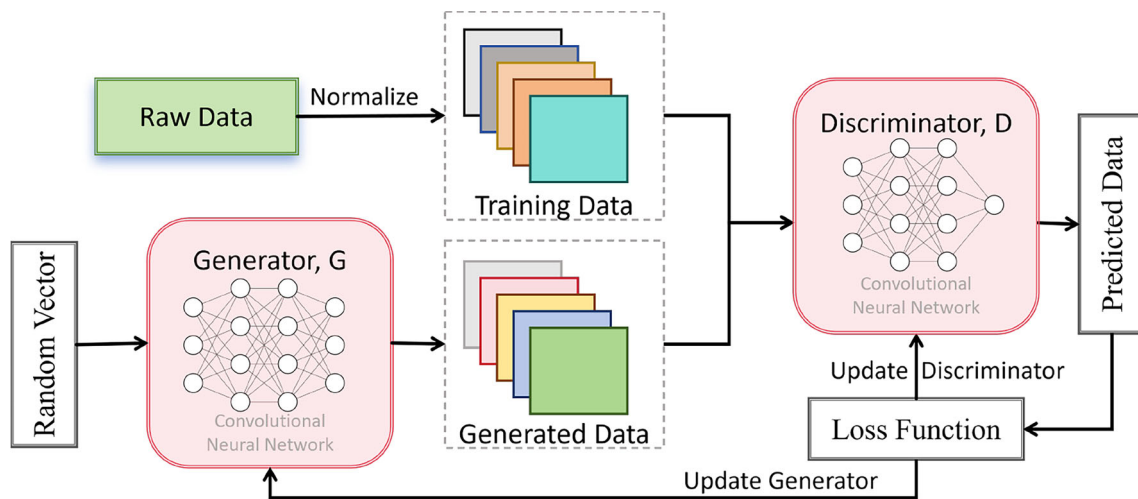


Fig. 2 The schematic diagram of GAN

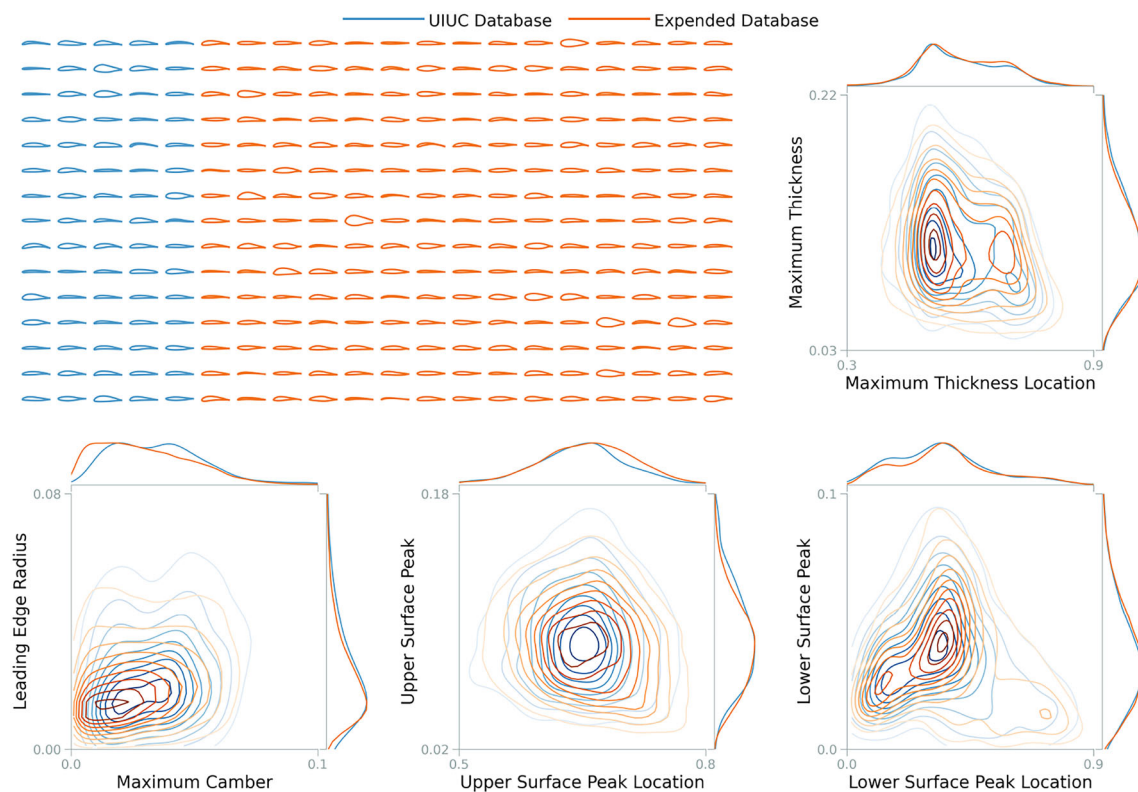


Fig. 3 Distributions of the training and generated airfoils

ingful gradient information when the two distributions have small probability distributions or do not completely overlap. Thus, WGAN is effective in avoiding the mode collapse issue. We use WGAN to learn the distribution of the UIUC airfoil database and then use the trained generator to create an augmented training database with 30,000 realistic airfoils. Some realistic airfoil shapes are shown in Fig. 3. The blue

airfoils are from the UIUC database, and the orange ones are from the augmented airfoil database. Distributions of two datasets are analyzed in terms of camber, thickness, and the peak positions of the upper and lower surfaces, revealing a strong similarity. These results demonstrate that the WGAN model effectively captures the underlying distribution of the training airfoils.

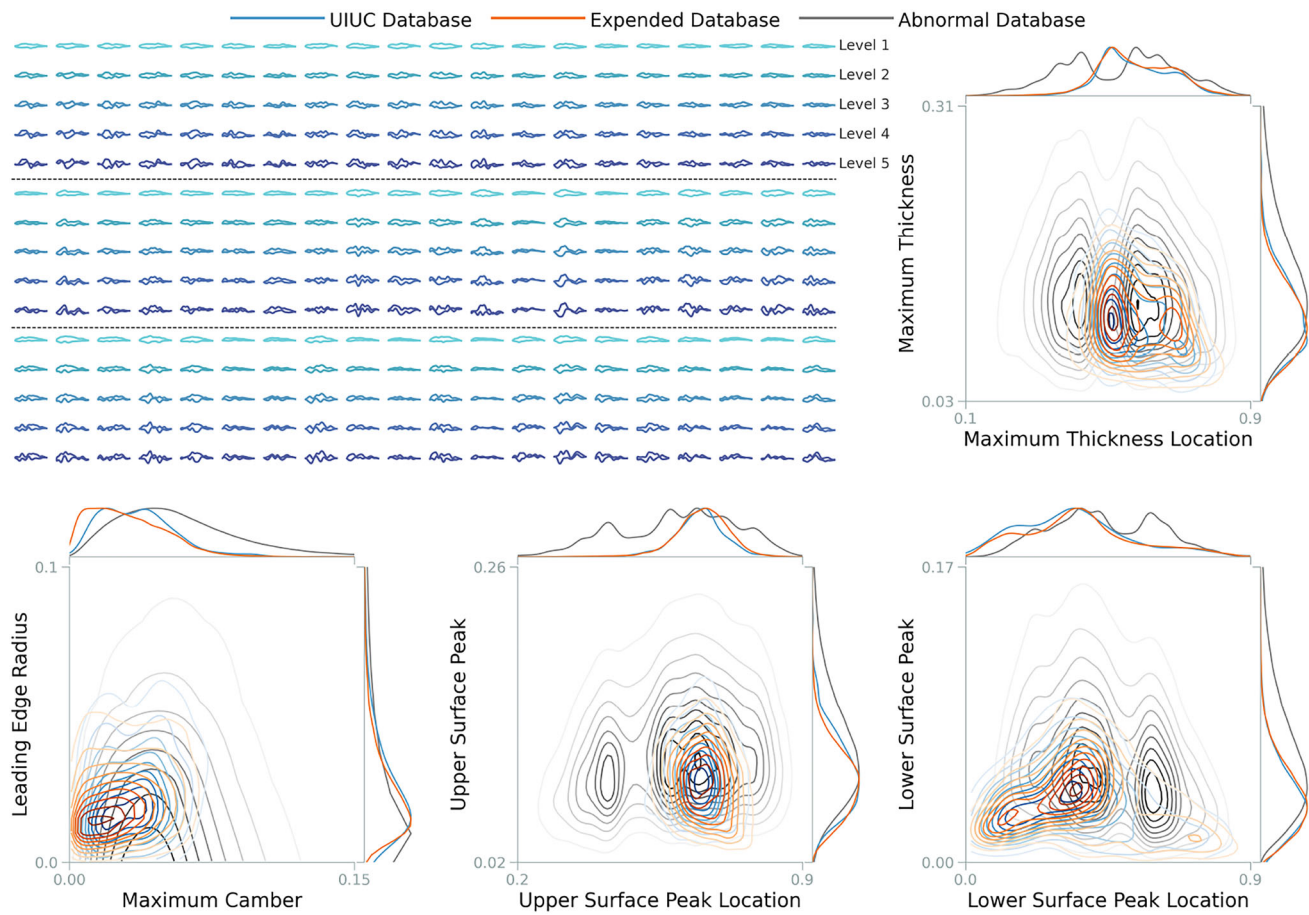


Fig. 4 Distributions of abnormal airfoils and realistic airfoils

2.2 Abnormal shape database construction

Training data of diverse abnormal airfoil shapes are also needed. These shapes can be obtained by harshly deforming realistic airfoils (Li et al. 2020). We use the FFD and the Latin hypercube sampling (LHS) method to generate abnormal airfoil shapes. The previous geometric validity model lacks generalizability, mainly due to the lack of diversity in the deformation magnitude. To address the issue, we build the database by involving abnormal airfoils with diverse perturbation magnitudes.

Five perturbation levels are used to generate abnormal airfoils. Bounds of the smallest level are defined as $\pm 40\%$ of the local thickness, and bounds of other levels increase linearly, i.e., $\pm 40\% \times n$ of local thickness for the n th level. Airfoil samples are generated in each bound level. For each level, ten sample shapes are generated around each UIUC airfoil, resulting in a total of $14,000 \times 5$ abnormal airfoils. Some sample airfoils are shown in Fig. 4, and their distribution is compared with UIUC and augmented realistic airfoils in terms of thickness, camber, and other geometric features. Abnormal airfoils extend into previously uncharted distribu-

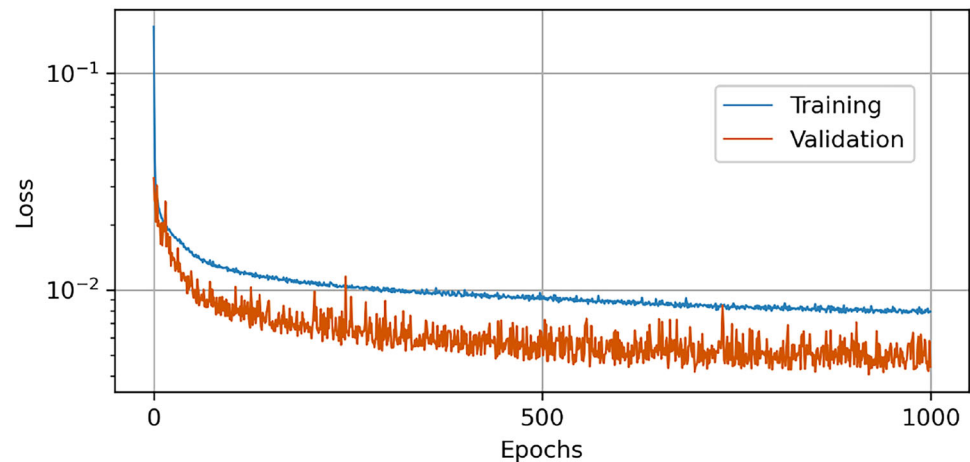
tion domains of UIUC and realistic airfoils, indicating the coverage of diverse geometric features. The abnormal diversity can enhance the capability of the data-driven validity model in identifying and filtering out unreasonable airfoil shapes, and its effectiveness will be demonstrated in the following section.

2.3 Deep-learning-based geometric validity model

We adopt the Convolutional Neural Network (CNN) to form the geometric validity analysis model. The network architecture consists of four convolutional layers. The first three layers have 64 filters with a filter size of 10, and the final layer has 128 filters to extract higher-level and more complex features. Between two convolutional layers, we incorporate a regularization layer to avoid overfitting and improve generalization capabilities.

The augmented realistic airfoil database and abnormal shape database introduced previously are used to train the geometric analysis model. All the realistic airfoils are labeled as 1. For the abnormal airfoils, we investigate two labeling strategies as follows. Firstly, all abnormal airfoils are set to

Fig. 5 Training loss convergence history for the geometric validity model



−1; Secondly, abnormal labels decrease from −0.6 to −1 with the increase of the perturbation magnitude. Specifically, abnormal airfoils with the first perturbation level are assigned a label value of −0.6, and the labels of subsequent levels decrease linearly to −1. We find the second labeling strategy enhances the generalizability of the validity model, and we adopt this approach to train the model in this paper. The loss function is defined as the mean squared error (MSE) between the predicted values and the given labels, that is,

$$\text{loss} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2. \quad (2)$$

As shown in Fig. 5, the network training converges smoothly.

As shown in Fig. 6, we compare the validity model developed in this paper with the model presented in previous work. Both models detect modest shape abnormalities in airfoils shown in Slice 3. The model developed in this paper exhibits much better performance in analyzing airfoils of Slices 1 and 2, where geometric abnormalities are more severe. The augmented dataset improves the model's capability of identifying various abnormalities. Also, validity scores vary continuously and smoothly, and this differentiable feature makes it feasible to be embedded in gradient-based optimization methods.

3 Hybrid aerodynamic shape optimization with geometric filtering

3.1 Drawbacks of directly coupling geometric filtering with gradient-based optimization

Gradient-based optimization, for example, using the Sequential Quadratic Programming (SQP) method, is efficient in solving high-dimensional nonlinear problems. Directly using the deep-learning-based geometric validity analysis model as

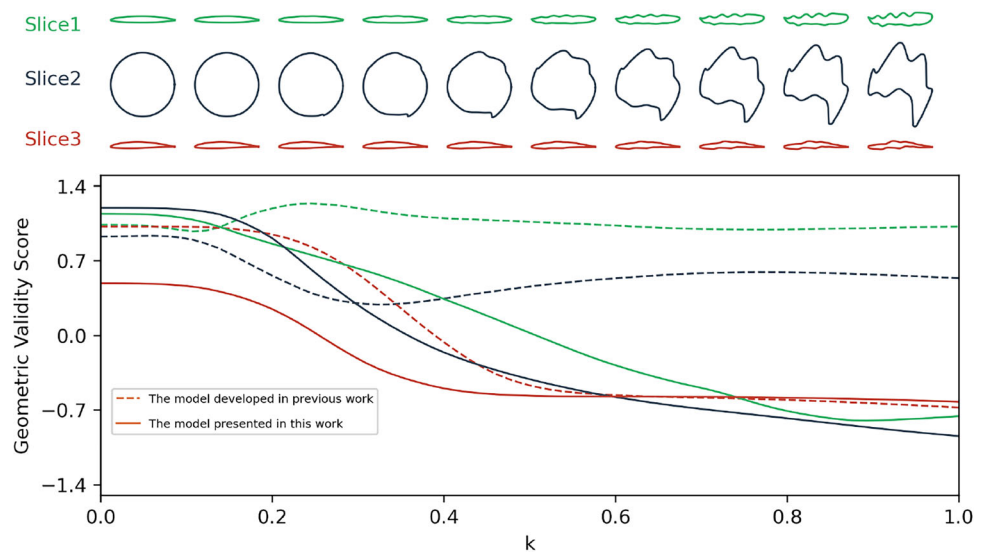
a constraint in gradient-based optimization can avoid useless searching of abnormal geometric domains and thus improve optimization efficiency. A preliminary study of this approach has shown to be effective in the drag reduction of a 2D cylinder shape (Li and Zhang 2021c). Nevertheless, we find this direct coupling approach ineffective in solving more challenging optimization problems, because it cannot ensure reasonable searching in line search processes.

We showcase the issue using a drag reduction case starting from a 3D cylinder shape in Fig. 7. Abnormal shapes are fed to the CFD solver in some line search processes, leading to failures of CFD mesh deformation or CFD simulations. This indicates that the geometric design space has not been effectively filtered using this direct coupling way, especially for line searching. More effective approaches to using geometric filtering models are required to improve optimization robustness.

3.2 Hybrid optimization approach

We propose a hybrid gradient-based aerodynamic shape optimization approach to effectively leverage deep-learning-based geometric filtering to enhance optimization efficiency and effectiveness. The benefits are twofold. Firstly, to reduce the optimization complexity, we decompose the optimization problem into a series of sub-optimization problems using aerodynamic approximations and trust regions with deep-learning-based geometric validity analyses. In each sub-optimization, the problem becomes more manageable and can be efficiently solved. Secondly, to avoid failures in line search processes, we develop a step size criterion using the deep-learning-based geometric validity model, called the *discriminative score criterion* (DSC). In gradient-based optimization algorithms with the backtracking strategy, such as the sequential least squares programming algorithm (SLSQP), the backtracking step size is usually fixed, and backtracked shapes are directly fed to CFD without any prior

Fig. 6 Geometric validity model developed in this work is generalizable to diverse shapes



validity analyses. We introduce DSC as an additional backtracking criterion that prevents the optimizer from involving aerodynamic shapes with low validity scores.

The hybrid optimization approach is composed of two phases. In Phase 1, we decompose the optimization into a sequence of sub-optimization problems subject to geometric validity constraints. Aerodynamic performances inside the trust region are approximated using aerodynamic sensitivities and the first-order Taylor expansion. In each sub-optimization, design variable bounds are constrained by the *trust region bound* (TRB), where the square sum of design variable changes cannot exceed the given TRB. Soft-constraint mechanisms are used to weaken the problem since constraints may be satisfied in the chosen region.

In Table 1, we use a drag reduction problem to showcase the sub-optimization settings. The original optimization is to reduce the drag with a lift constraint ($C_l \geq C_l^{\text{con}}$), and the design variables include n_{FFD} shape variables and the angle of attack. C_d^0 and C_l^0 are the drag and lift coefficients of the sub-optimization baseline shape, which are accurately solved using the CFD simulation. g_d^i and g_l^i are the derivatives of the drag and lift coefficients with respect to each shape design variable, respectively, which are solved using the adjoint method. In first a few sub-optimizations, aerodynamic constraints may not be satisfied due to the TRB constraint. We use relaxed constraints that are progressively tightened throughout the optimization process. For example, the lift constraint is set as $C_l \geq 0$ for the initial circular shape and gradually moves toward $C_l \geq 0.824$ as the optimization advances. After one sub-optimization is solved, the optimized shape is used as the baseline for the next sub-optimization.

We terminate Phase 1 when the convergence criterion (ϵ) is met. ϵ is defined as the ratio of the difference between the

average objective function value over the last n steps and the minimum value, relative to the difference between the initial value and the minimum value, given by:

$$\epsilon = \frac{\frac{1}{n} \sum_{i=N-n}^N f_{\text{objective}}^i - f_{\text{objective}}^{\min}}{f_{\text{objective}}^0 - f_{\text{objective}}^{\min}} \times 100\%. \quad (3)$$

Here, N refers to the total number of steps computed at that time, and $f_{\text{objective}}^{\min}$ represents the minimum $f_{\text{objective}}$ so far.

In Phase 2, we solve the optimization starting from the shape obtained in Phase 1. As shown in Algorithm 1, we modify a conventional gradient-based optimization algorithm (the SLSQP algorithm) using the deep-learning-based geometric validity model. The algorithm follows the classical SLSQP framework, using the Han–Powell quasi-Newton method with a BFGS update of the B-matrix and an L1-test function in the step-length algorithm. The line search is modified with geometric filtering. We introduce a backtracking criterion using the deep-learning-based geometric filter, where the line search step size is directly reduced when the geometric abnormality exceeds the given constraint. We allow a maximum of ten backtracking attempts to avoid excessive operations. This mechanism helps avoid spending excessive computational costs on unfavorable results.

The XDMS (eXtended Design Structure Matrix) diagram of the proposed hybrid optimization method is shown in Fig. 8. The optimization settings module is responsible for distinguishing the optimization phase, defining design variable bounds, and selecting aerodynamic evaluation functions. The primary motivation of the proposed method is to improve optimization efficiency and robustness by constraining invalid regions in the design space. We demonstrate the effectiveness of this method using two challenging cases (a

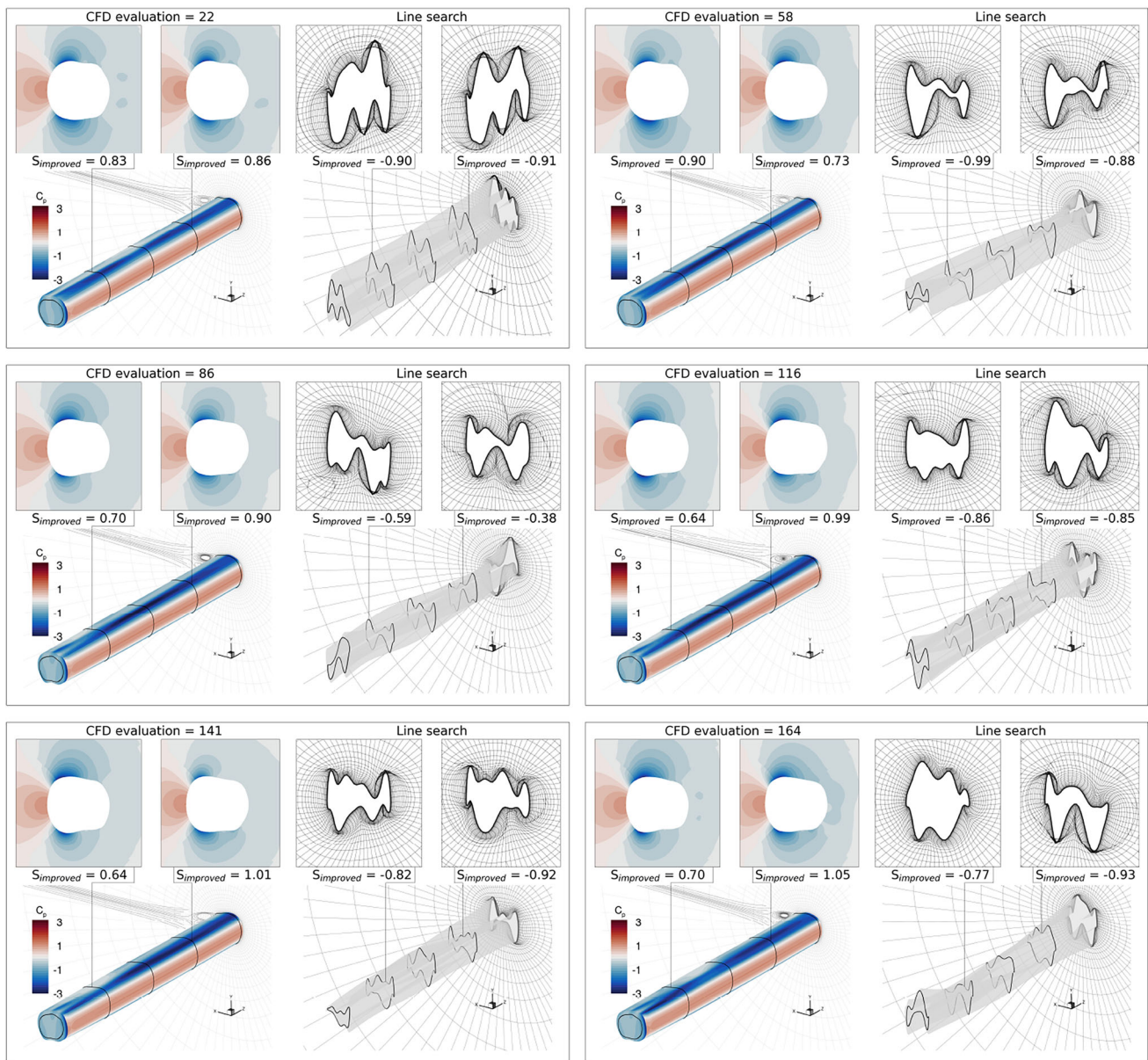


Fig. 7 Directly coupling geometric filtering with gradient-based optimization fails

Table 1 Optimization problem statement in Phase 1 of the hybrid optimization approach

	Functions	Quantity	Description
Minimize	C_d	1	Approximate drag coefficient $C_d = C_d^0 + \sum_{i=1}^n g_d^i x_{\text{shape}}^i + g_d^\alpha \alpha$
With respect to	α	1	Angle of attack
	$\mathbf{x}_{\text{shape}}$	n_{FFD}	Perturbation of FFD control points
Subject to	$C_l \geq C_l^{\text{con}}$	1	Approximate lift coefficient $C_l = C_l^0 + \sum_{i=1}^n g_l^i x_{\text{shape}}^i + g_l^\alpha \alpha$
	$\sigma \leq \text{TRB}$	1	$\sigma = \alpha^2 + \sum_{i=1}^{n_{\text{point}}} (x_{\text{shape}}^i)^2$
	$S \geq \text{DSC}$	1	Geometric validity constraint

2D case starting from a circle and a 3D case starting from a cylinder) in the following sections.

Algorithm 1 Modified SLSQP algorithm with deep-learning-based geometric filtering

Input: Objective $f(\mathbf{x})$, constraints $g(\mathbf{x}) \leq 0$, $h(\mathbf{x}) = 0$,
Initial design $\mathbf{x}_0$, tolerance δ , maximum iterations $k_{\max}$,
Initial step α_0 , line search factor c ,
Geometric filter **GEO**($\cdot$), discriminative score criterion

DSC
Output: Optimized design variables $\mathbf{x}^*$

```

1: procedure MODIFIEDSLSQP( $\mathbf{x}_0, f, g, h, \delta, \alpha_0, k_{\max}, \mathbf{GEO}, \mathbf{DSC}$ )
2:   Initialize  $\mathbf{x} \leftarrow \mathbf{x}_0, k \leftarrow 0$ 
3:   Initialize Hessian approximation  $\mathbf{B}$  of the Lagrangian function
    $L(\mathbf{x}, \lambda, \mu)$ 
4:   while not converged and  $k < k_{\max}$  do
5:     Evaluate  $f(\mathbf{x}), \nabla f(\mathbf{x})$ 
6:     Evaluate  $g(\mathbf{x}), h(\mathbf{x}), \nabla g(\mathbf{x}), \nabla h(\mathbf{x})$ 
7:     Construct QP subproblem (linearized constraints):

       
$$\begin{aligned} \min_{\mathbf{p}} \quad & \frac{1}{2} \mathbf{p}^T \mathbf{B} \mathbf{p} + \nabla f(\mathbf{x})^T \mathbf{p} \\ \text{s.t.} \quad & g(\mathbf{x}) + \nabla g(\mathbf{x})^T \mathbf{p} \geq 0 \\ & h(\mathbf{x}) + \nabla h(\mathbf{x})^T \mathbf{p} = 0 \end{aligned}$$


8:     Construct the corresponding Lagrangian function:

       
$$L(\mathbf{x}, \lambda, \mu) = f(\mathbf{x}) - \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) - \sum_{j=1}^n \mu_j h_j(\mathbf{x})$$


9:     Solve QP subproblem to obtain search direction  $\mathbf{p}$  with  $\mathbf{B}$ 
10:    Set initial step size  $\alpha \leftarrow \alpha_0$ 
11:    for  $i = 1$  to 10 do
12:      if  $\mathbf{GEO}(\mathbf{x} + \alpha \mathbf{p}) > \mathbf{DSC}$  then
13:        Evaluate L1 merit function:

          
$$\begin{aligned} \Phi(\mathbf{x}, \lambda, \mu; \varrho) = & f(\mathbf{x}) - \sum_{j=1}^n \left( \mu_j h_j(\mathbf{x}) - \frac{1}{2} \varrho_j h_j(\mathbf{x})^2 \right) \\ & - \sum_{i=1}^m \begin{cases} \lambda_i g_i(\mathbf{x}) - \frac{1}{2} \varrho_i g_i(\mathbf{x})^2, & g_i(\mathbf{x}) \leq \frac{\lambda_i}{\varrho_i} \\ \frac{1}{2} \frac{\lambda_i^2}{\varrho_i}, & \text{otherwise} \end{cases} \end{aligned}$$


14:        if  $\Phi(\mathbf{x} + \alpha \mathbf{p}) \leq \Phi(\mathbf{x}) + c\alpha \nabla f(\mathbf{x})^T \mathbf{p}$  then
15:           $\mathbf{x} \leftarrow \mathbf{x} + \alpha \mathbf{p}$ 
16:          break
17:        else
18:           $\alpha \leftarrow 0.1 \cdot \alpha$ 
19:        end if
20:      else
21:         $\alpha \leftarrow 0.1 \cdot \alpha$ 
22:      end if
23:      if  $i = 10$  then
24:        (Fallback)  $\mathbf{x} \leftarrow \mathbf{x} + \alpha \mathbf{p}$ 
25:      end if
26:    end for
27:    Update  $\mathbf{B}$  using BFGS update formula
28:    if  $\|\nabla f(\mathbf{x}) + \lambda^T \nabla g(\mathbf{x}) + \mu^T \nabla h(\mathbf{x})\| < \delta$  then
29:      converged
30:    else
31:       $k \leftarrow k + 1$ 
32:    end if
33:  end while
34:  return  $\mathbf{x}^* \leftarrow \mathbf{x}$ 
35: end procedure

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4 Two-dimensional aerodynamic shape optimization: from a circle to an airfoil

4.1 Problem description

We show the effectiveness of the proposed hybrid optimization approach in an aerodynamic shape optimization problem starting from a 2D circle. The optimization is to reduce the drag coefficient, subject to constraints on the lift coefficient and the shape area. Details of this optimization problem are shown in Table 2.

For CFD simulations and adjoint evaluations, we use the open-source ADflow solver (Mader et al. 2020; Kenway et al. 2019; Yildirim et al. 2019). The FFD method implemented in pyGEO (Hajdik et al. 2023) is used for shape parameterization and deformation. As shown in Fig. 9, 20 control points are used as shape design variables. The distribution of FFD control points is not homogeneous, and the points near the leading and trailing edges are relatively denser. This distribution is designed to allow for more detailed shape deformation capability in these critical regions. Since we add the angle of attack as the independent design variable, the FFD control points at the leading and trailing edges are constrained to move in equal and opposite directions to avoid shearing twists. Gradient-based optimization algorithms (such as SLSQP) implemented in pyOptSparse (Wu et al. 2020) are used. The optimization is performed with flow conditions of $M = 0.734$ and $Re = 5 \times 10^6$. The first layer mesh thickness is set to 5×10^{-6} m, which corresponds to $y^+ \approx 1$, to ensure simulation fidelity.

4.2 Optimization results

We first evaluate conventional gradient-based optimization. Two optimization algorithms, SLSQP and the Interior Point Optimizer (IPOPT), are applied without geometric validity constraints. A third approach directly integrates geometric filtering with SLSQP, enforcing a validity constraint ($S \geq 0.9$). Figure 10 shows convergence histories (drag coefficients and shape evolutions) for all three methods. Drag coefficient values are shown in counts (1 y-axis unit = 100 counts, and one count corresponds to 0.0001 drag coefficient). All optimizations exhibit oscillatory convergence, failing at abnormal shapes after tens of iterations. These results demonstrate that conventional gradient-based optimization cannot effectively handle large geometric design spaces, regardless of the optimization algorithm choice (SLSQP or IPOPT). Also, direct integration of geometric filtering merely provides marginal improvements in design space exploration and remains insufficient for convergence.

We then use the proposed hybrid optimization method to solve this problem. The optimization converges smoothly with a cost of 268 CFD evaluations, and the drag coeffi-

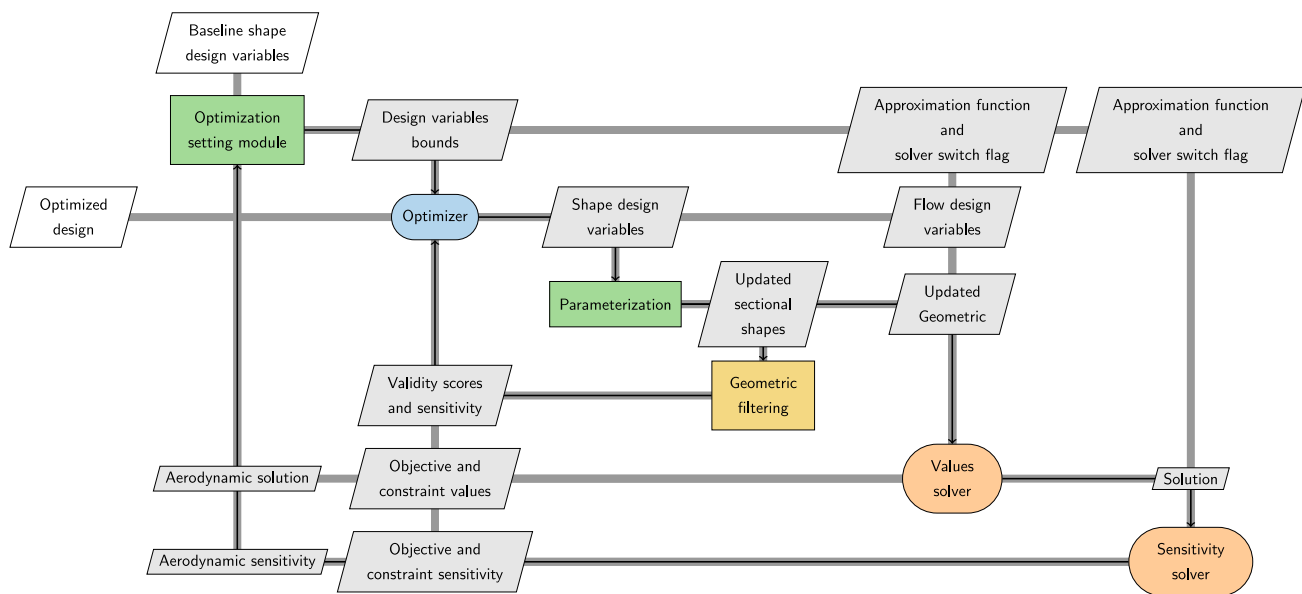
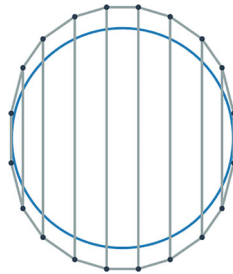


Fig. 8 XDSM diagram of the hybrid optimization method

Table 2 Optimization problem statement of the 2D circle case

	Functions	Quantity	Description
Minimize	C_d	1	Drag coefficient
With respect to	α	1	Angle of attack
	$\mathbf{y}$	20	$\mathbf{y}$ Perturbations of FFD control points
Subject to	$C_l = 0.824$	1	Lift constraint
	$A \geq 0.08 \times A_{\text{initial}}$	1	Area constraint
	$S \geq \text{DSC}$	1	Geometric validity constraints

Fig. 9 FFD control box used in the 2D circle case



cient and the shape evolution history are shown in Fig. 11. In Phase 1, we use a thrust region criterion of $\text{TRB} = 0.005$ and a termination criterion of $\epsilon = 10\%$. With this setting, Phase 1 involves 91 CFD evaluations, and a shape with a discriminative score of 0.9 is achieved with the drag coefficient reduced by 88.1%. In Phase 2, the optimization continues from the shape solved in Phase 1, and the deep-learning-based discriminative score criterion is set as $\text{DSC} = 0.5$. The final drag coefficient of the optimized shape is 128.21 counts, representing a 98.8% reduction compared to the initial drag coefficient of the circle. These results highlight the efficiency and effectiveness of the proposed hybrid optimization method.

To further demonstrate the robustness of the proposed hybrid method, we present optimization results with different settings. Figure 12 shows the impact of changes in TRB and ϵ on the optimization process. As shown in the first row of Fig. 12, the termination criterion of the first phase ϵ does not significantly affect the optimization convergence, and it can be chosen from a wide range (1–20%). This is because the improved gradient-based optimization in Phase 2 shows a significantly high convergence capability and makes the optimization effectively converge even with a less-than-ideal starting shape. However, as shown in the second row of Fig. 12, we find that an excessively large TRB can lead to inaccuracies in the first-order Taylor expansion, causing the optimizer to deviate from the correct direction and ultimately leading to failed results. This highlights the importance of selecting appropriate values for TRB to achieve reliable and effective optimization outcomes. Generally, the proposed optimization method shows adequate robustness in parameter settings.

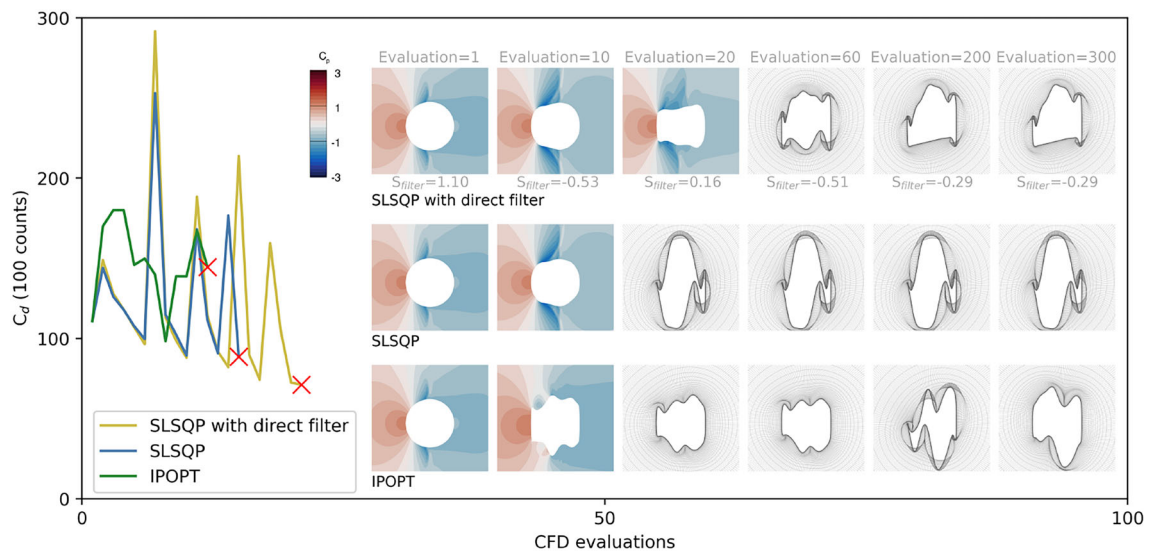


Fig. 10 Conventional gradient-based optimization fails in the 2D cylinder case

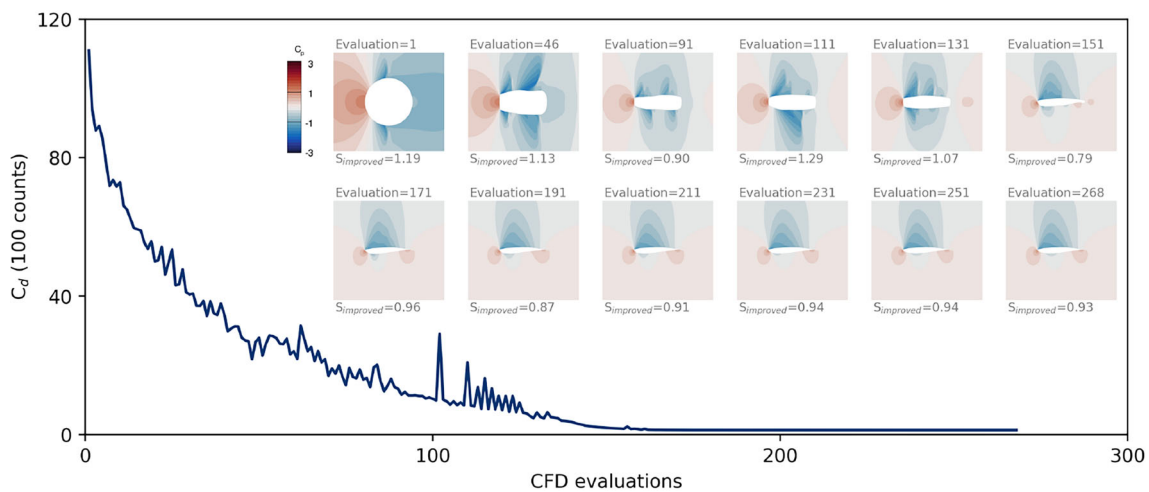


Fig. 11 The proposed hybrid optimization method effectively converges to an airfoil in the 2D case

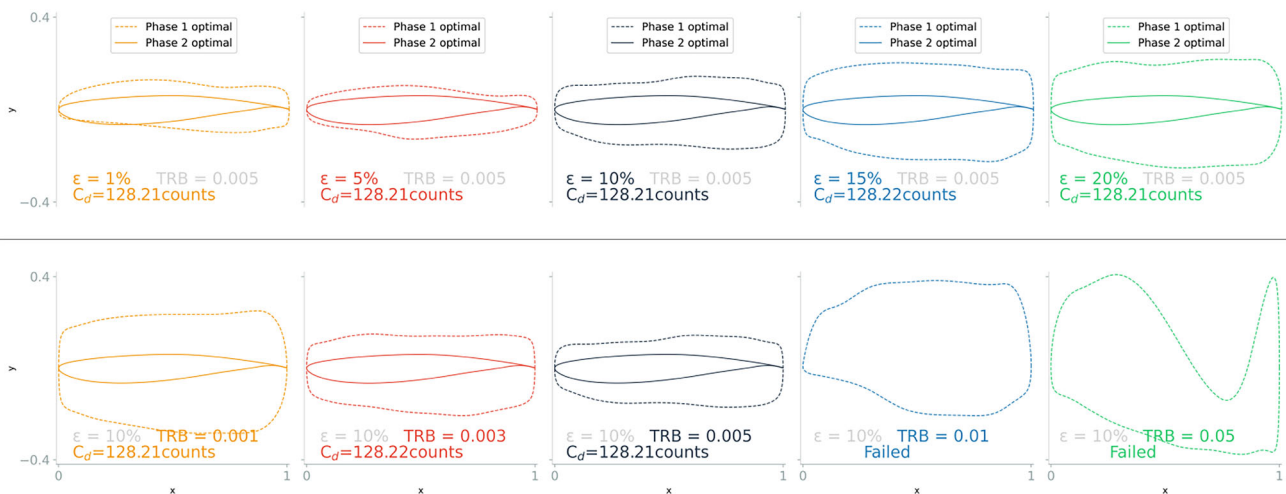
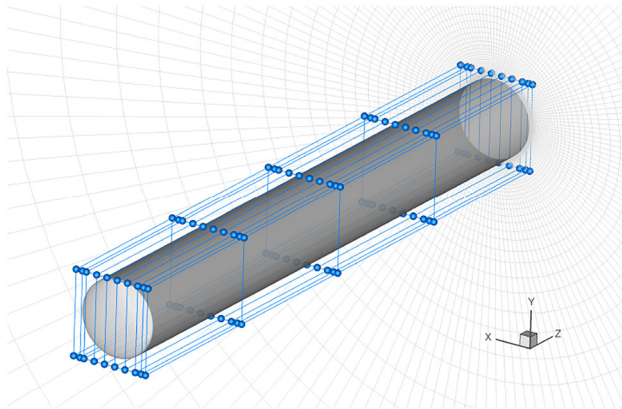


Fig. 12 Aerodynamic shape optimization is robust subject to a reasonably chosen TRB

Table 3 Optimization problem statement of the 3D cylinder case

	Functions	Quantity	Description
Minimize	C_D	1	Drag coefficient
With respect to	α	1	Angle of attack
	$\mathbf{y}$	100	$\mathbf{y}$ Perturbation of FFD control points
Subject to	$C_L = 0.8$	1	Lift constraint
	$S \geq \text{DSC}$	5	Geometric validity constraints
	$T \geq T_0$	500	Thickness constraint

**Fig. 13** CFD mesh and FFD control box of the 3D cylinder

5 Three-dimensional aerodynamic shape optimization: from a cylinder to a wing

5.1 Problem description

We further validate the proposed hybrid optimization method in a more challenging optimization case—drag reduction of a 3D cylinder shape subject to lift and thickness constraints. Details of this optimization case are listed in Table 3. FFD with 100 control points is used to parameterize the cylinder shape. An equality constraint on the lift coefficient is posed ($C_L = 0.8$). 500 thickness constraints are posed to ensure that the optimized shape has sufficient volume. Geometric validity constraints are imposed on sections at 1%, 25%, 50%, 75%, and 99.4% of the wingspan. Monitor shapes compatible with the validity model are defined at these positions and are deformed together with the CFD mesh during optimization. This enables efficient detection of geometric abnormalities across any sections for both structured and unstructured meshes. The discriminative score criterion is set as $\text{DSC} = 0.5$. The CFD mesh, the FFD control box, and the initial shape are shown in Fig. 13. The optimization is performed with flow conditions of $M = 0.3$ and $Re = 4.35 \times 10^6$.

5.2 Optimization results

We compare differences between adjoint-based optimization with the direct geometric filtering approach and the proposed hybrid optimization method. As shown in Fig. 14, adjoint-based optimization with direct geometric filtering fails even after nearly 200 CFD simulations. The iterative optimization shapes are shown in Fig. 15, which shows that direct filtering using the validity analysis model does not work as expected. Many abnormal shapes are fed to the CFD solver, and some of them fail at the mesh deformation process. The results are consistent with the investigation in the 2D circle case, highlighting the limitations of the direct geometric filtering approach. As shown in Fig. 16, the proposed hybrid optimization method converges smoothly in this challenging case. No abnormal wing shapes are fed to CFD, showing an effective geometric filtering capability. The hybrid optimization method uses 622 CFD evaluations in total, with 82 CFD evaluations used in Phase 1. The drag coefficient reduces by 68.6% from 4367.54 counts to 1372.87 counts in Phase 1. The final drag coefficient of the optimal shape in Phase 2 is 239.94 counts, which is 94.5% lower than the initial drag coefficient. After the optimization, the cylinder is optimized to a wing shape, with all sectional shapes to be typical subsonic airfoils. These results highlight the capability of the proposed hybrid optimization method in solving challenging aerodynamic shape optimization problems.

We further investigate the robustness of the proposed optimization method in terms of parameter settings. In Fig. 17, we show the influence of ϵ on the optimization result. Similarly to the 2D case, the impact of ϵ on the optimization is slight, and all cases with ϵ increasing from 1 to 20% converge to the same optimal shape. Different from the 2D case, the ability of Phase 1 to approach the optimal geometric sub-domain diminishes. Nevertheless, the modified gradient-based optimization algorithm used in Phase 2 still enables successful convergence. TRB impacts the optimization convergence. When it is too large, the aerodynamic approximations are not reliable, and thus, Phase 1 may fail to find a reasonable shape. As shown in Fig. 18, an excessively large TRB leads to optimization failure. Thus, to ensure optimization convergence, TRB should be chosen reasonably.

Fig. 14 Direct geometric filtering has worse performance when solving more challenging problems

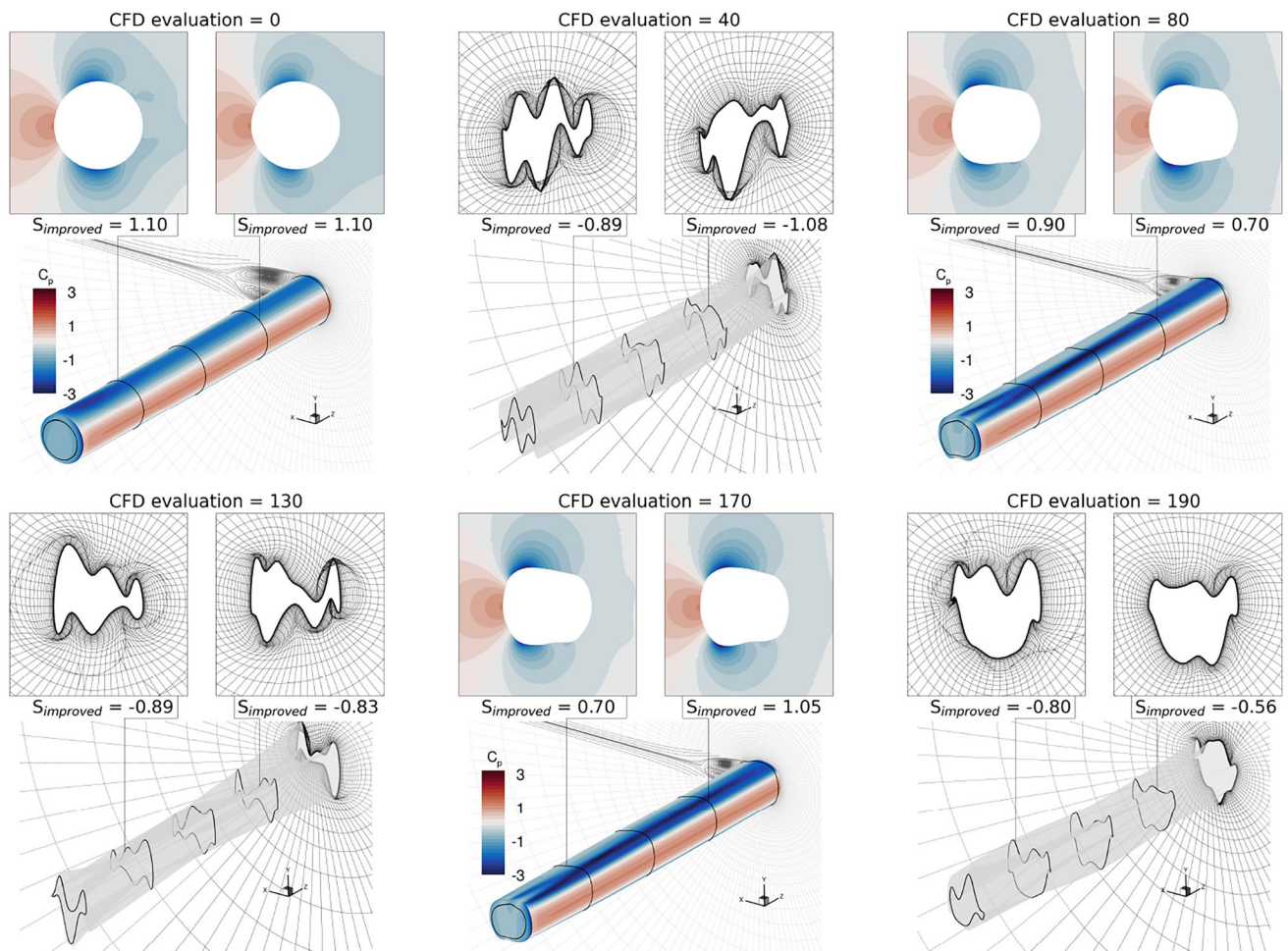
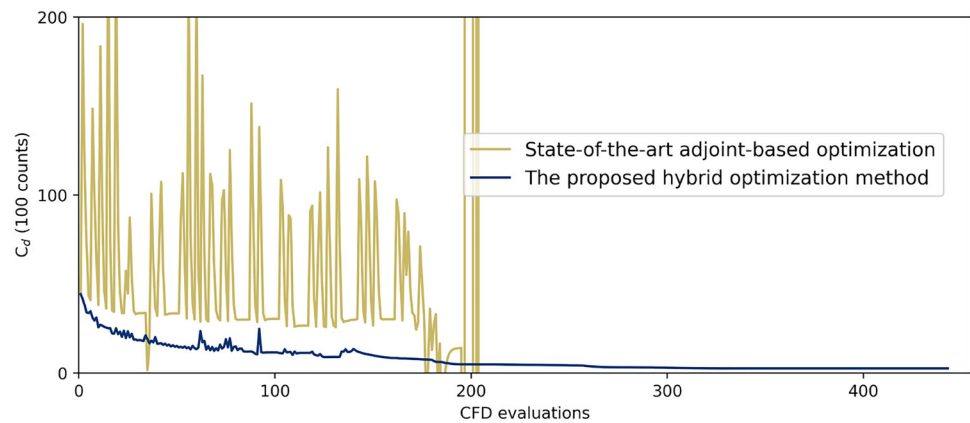


Fig. 15 State-of-the-art adjoint-based optimization fails in solving the 3D cylinder problem

This work aims to improve ASO robustness via effectively leveraging historical designs and embedding the data-driven model with the optimization algorithm. Case studies starting from bluff geometries (circle and cylinder) are chosen to demonstrate the method's effectiveness in extreme scenarios. For typical ASO problems, well-designed baseline shapes are often available, likely reducing optimization difficulty to a

manageable level for conventional gradient-based methods. However, optimization failures remain a significant challenge, particularly when the number of design variables or their bounds are not carefully selected. The proposed hybrid optimization method effectively addresses this issue as well.

To illustrate this point, we examine a wing shape optimization problem with NACA0012 airfoil-based baseline

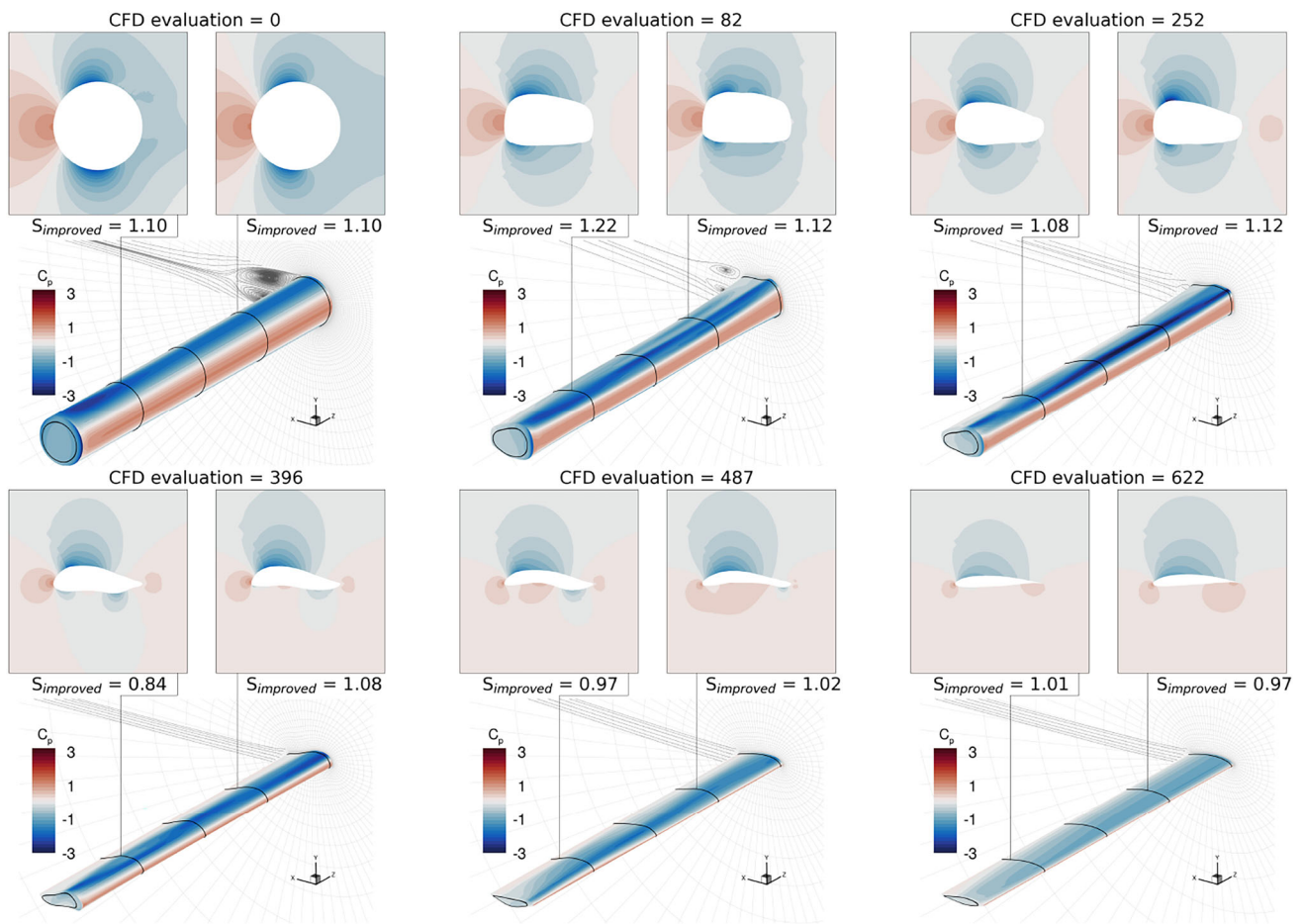


Fig. 16 3D cylinder shape is optimized to a subsonic wing using the proposed hybrid method

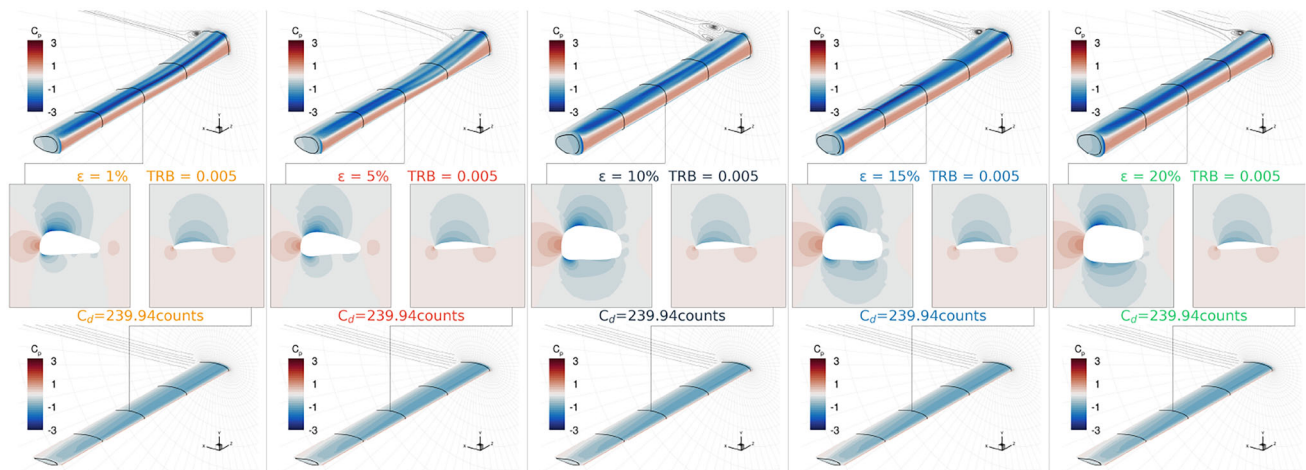


Fig. 17 The proposed hybrid optimization method is robust with different ϵ settings

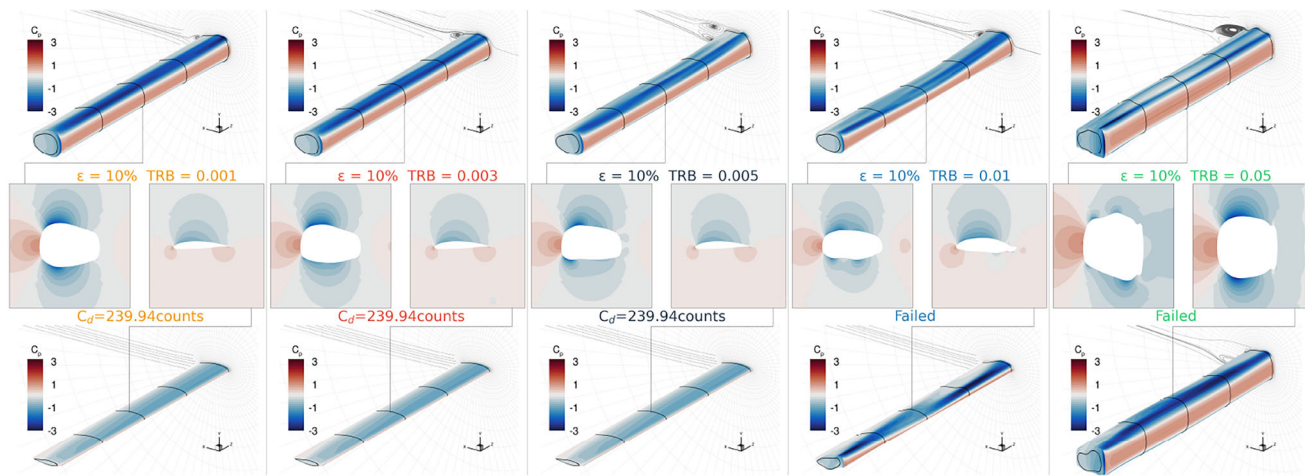


Fig. 18 The proposed hybrid optimization method is robust subject to a reasonable TRB

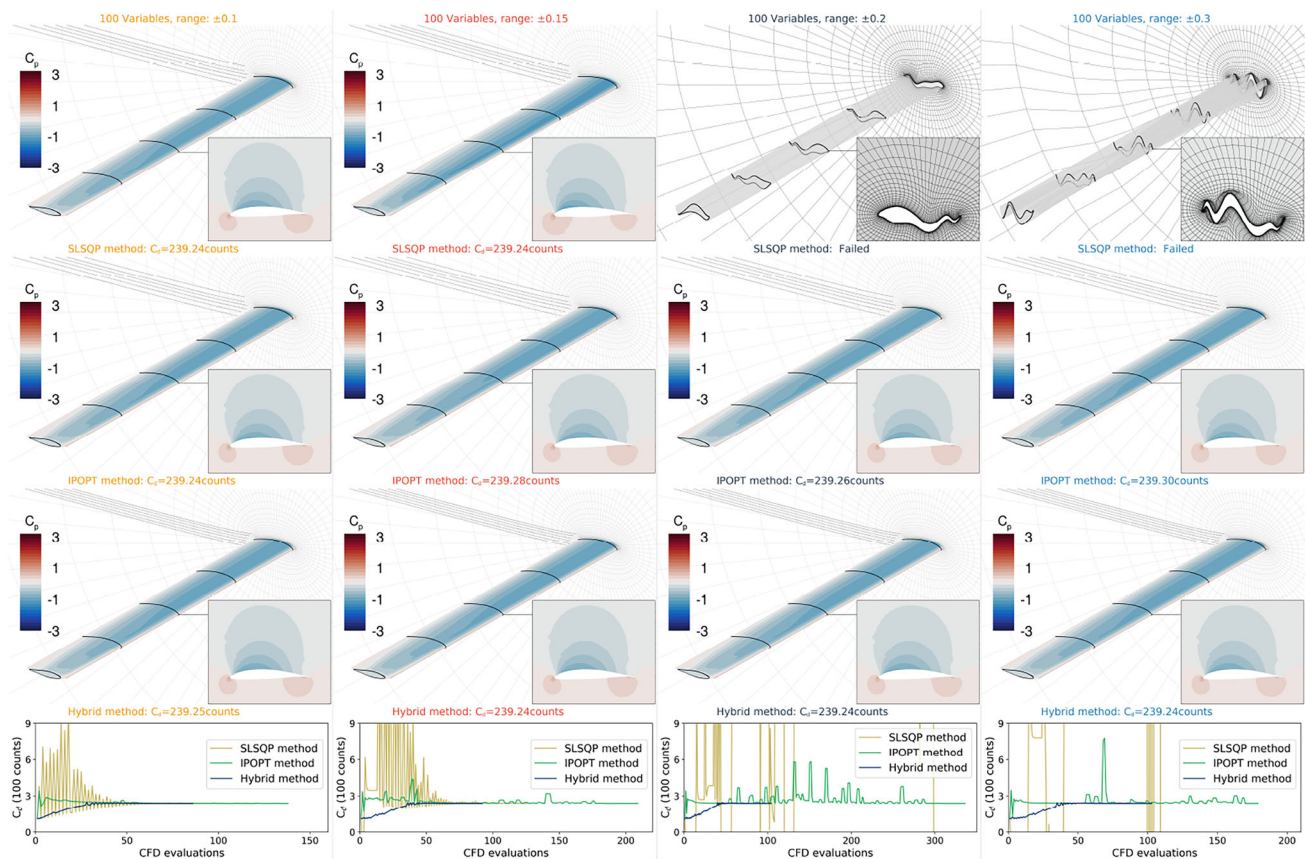


Fig. 19 The hybrid optimization method is robust against the increase of design variable bounds

sections. The wingspan, objective function, and constraints are the same as those of the cylinder optimization case. First, we show the impact of variable bounds when the number of design variables is 100. A comparison is conducted between conventional optimization algorithms (SLSQP and IPOPT) and the proposed hybrid optimization method. As shown in Fig. 19, SLSQP performs well with moderate variable bounds

but fails to converge when the bounds exceed 0.2. IPOPT shows stronger robustness and maintains convergence even under large variable bounds. SLSQP requires about 120 CFD iterations in converged cases, but it lacks robustness in challenging cases. IPOPT offers superior robustness at a higher computational cost than SLSQP. The proposed hybrid method is both the most efficient and robust, converging reli-

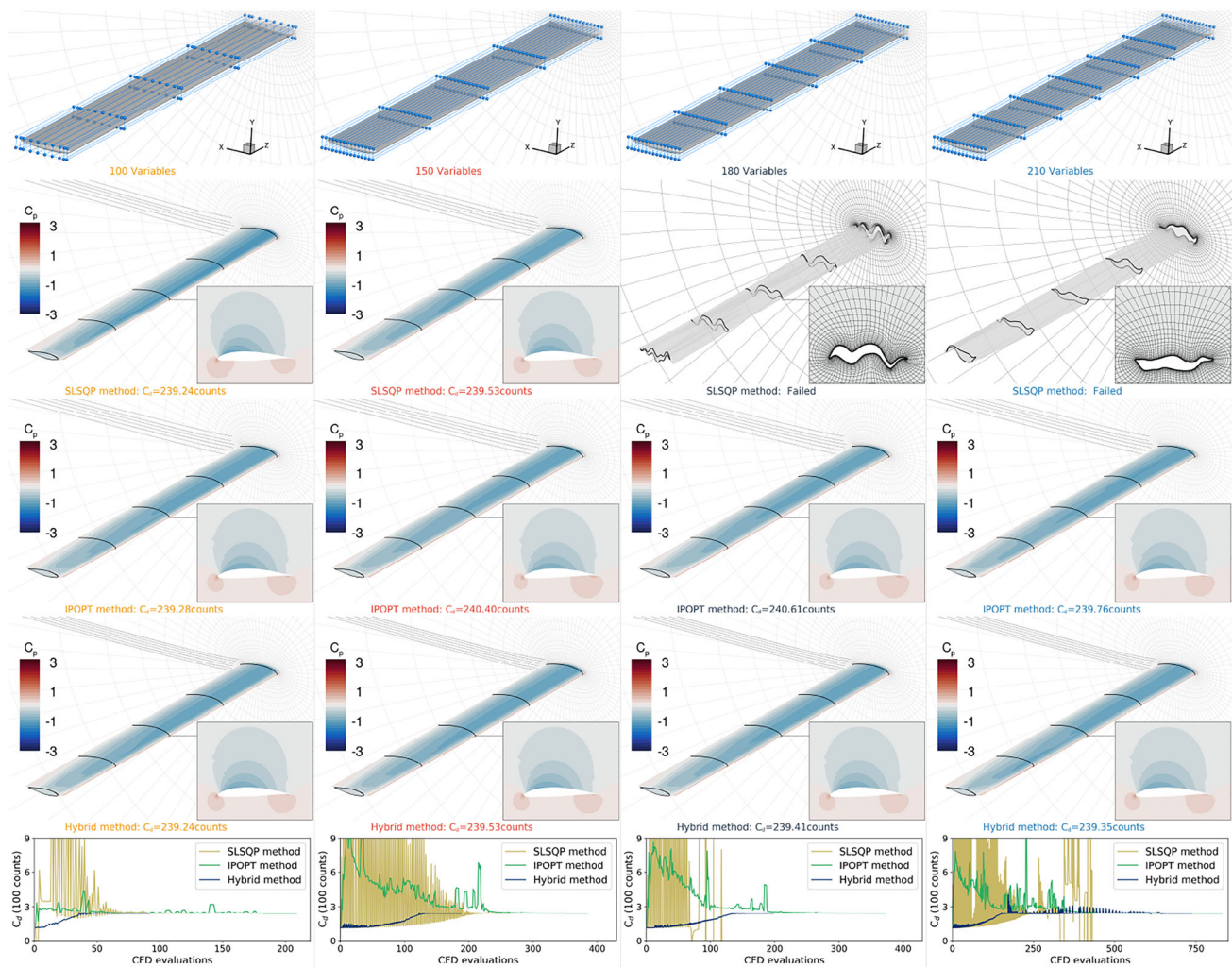


Fig. 20 The hybrid optimization method is robust against the increase of design variables

ably in just 80–110 CFD iterations. Then, we investigate the influence of design variable numbers on optimization convergence (Fig. 20). SLSQP again fails to converge beyond a certain number of design variables. IPOPT still converges up to 210 design variables, with a cost of about 350 CFD iterations in all but one case. The choice of optimizers has a significant influence in adjoint-based optimization. The proposed hybrid method consistently converges to the optimal solution with the highest efficiency, requiring approximately 200 CFD iterations in most cases.

As discussed by Martins (2022), the convergence of adjoint-based ASO depends on multiple factors including mesh deformation, CFD solver reliability, derivative computation accuracy, and optimizer characteristics. Conventional gradient-based optimization imposes stringent robustness requirements on these components, and even with a well-designed baseline shape, optimization success is not always guaranteed. Our proposed method addresses these challenges by leveraging deep learning to eliminate the need

for time-consuming manual tuning of geometric design parameters. This reduces the robustness burden on the key ASO components, making robust optimization more accessible. However, since the data-driven geometric filtering models rely on historical designs (e.g., UIUC airfoils), the approach is not universally applicable. For design tasks lacking sufficient historical data, this strategy has limitations, necessitating alternative solutions.

6 Conclusion

We propose a deep-learning geometric filtering based hybrid aerodynamic shape optimization method to enhance optimization efficiency and robustness in solving challenging ASO problems. We first present a more generalizable geometric abnormality analysis model by developing enhanced airfoil databases. The model shows significant generalizability in abnormality analyses and exhibits a stronger capability

of effective geometric filtering, which forms the foundation of the hybrid optimization method. Then, we develop the hybrid optimization method based on the trust region strategy, deep-learning-based geometric filtering, and a modified gradient-based optimization algorithm. The hybrid method contains two phases. In the first phase, the optimization is addressed by solving a series of sub-optimization problems, where aerodynamic coefficients are approximated using aerodynamic sensitivities and the first-order Taylor expansion. Geometric validity constraints are imposed in these sub-optimization problems. In the second phase, the optimization problem is accurately solved using a modified gradient-based optimization algorithm. The modification of the optimization algorithm lies in constraining the line search step using strict geometric validity constraints. The hybrid method is designed to effectively leverage deep-learning-based geometric filtering to enhance optimization efficiency.

We demonstrate the performance of the proposed method in two challenging ASO cases, reducing the drag of 2D circle and 3D cylinder shapes, respectively. State-of-the-art adjoint-based optimization does not converge in both cases. However, the proposed method smoothly solves both problems, optimizing the bluff shapes to an airfoil and a wing, respectively, with significant drag reductions. Optimization with different parameter settings is also investigated, and the results show that the proposed method is robust as long as a reasonable trust region bound is used in the first phase. Furthermore, comparisons in more practical scenarios show that the hybrid method consistently preserves superior robustness and efficiency over conventional gradient-based optimization. Via effectively using deep-learning-based geometric validity analyses, the proposed method enables solving challenging aerodynamic shape optimization problems in a large and high-dimensional design space, hence promoting the findings of more efficient aerodynamic designs.

Author Contributions Yan Chen: Conceptualization, methodology, validation, and writing—original draft. Jichao Li: Conceptualization, methodology, and writing—review, editing, and supervision. Chao Song: Methodology and writing—review. Hongyang Liu: Methodology and writing—review. Zhu Zhou: Conceptualization and writing—review. Jiangtao Huang: Conceptualization and writing—review. Jinsheng Cai: Conceptualization and writing—review. B.C. Khoo: Conceptualization and writing—review.

Data Availability The data generated and optimized in this paper are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Replication of Results The models developed in this paper are available from the corresponding author upon reasonable request. Results can

be replicated by adopting the presented optimization method with the provided parameters.

References

- Arjovsky M, Chintala S, Bottou L (2017) Wasserstein GAN. arXiv preprint. [arXiv:1701.07875](https://arxiv.org/abs/1701.07875)
- Bartoli N, Lefebvre T, Dubreuil S, Olivanti R, Priem R, Bons N, Martins JRRA, Morlier J (2019) Adaptive modeling strategy for constrained global optimization with application to aerodynamic wing design. *Aerosp Sci Technol* 90:85–102. <https://doi.org/10.1016/j.ast.2019.03.041>
- Bons NP, He X, Mader CA, Martins JRRA (2019) Multimodality in aerodynamic wing design optimization. *AIAA J* 57(3):1004–1018. <https://doi.org/10.2514/1.j057294>
- Chai X, Yu X, Wang Y (2018) Multipoint optimization on fuel efficiency in conceptual design of wide-body aircraft. *Chin J Aeronaut* 31(1):99–106. <https://doi.org/10.1016/j.cja.2017.10.006>
- Chen W, Ramamurthy A (2021) Deep generative model for efficient 3D airfoil parameterization and generation. In: *AIAA Scitech 2021 Forum, VIRTUAL EVENT*, 2021. American Institute of Aeronautics and Astronautics. <https://doi.org/10.2514/6.2021-1690>
- Chen W, Chiu K, Fuge M (2019) Aerodynamic design optimization and shape exploration using generative adversarial networks. In: *AIAA Scitech 2019 Forum*, 2019. American Institute of Aeronautics and Astronautics, San Diego. <https://doi.org/10.2514/6.2019-2351>
- Du X, He P, Martins JRRA (2021) Rapid airfoil design optimization via neural networks-based parameterization and surrogate modeling. *Aerosp Sci Technol* 113:106701. <https://doi.org/10.1016/j.ast.2021.106701>
- Duan Y, Wu W, Zhang P, Tong F, Fan Z, Zhou G, Luo J (2019) Performance improvement of optimization solutions by pod-based data mining. *Chin J Aeronaut* 32(4):826–838. <https://doi.org/10.1016/j.cja.2019.01.014>
- Goodfellow I, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, Courville A, Bengio Y (2014) Generative adversarial nets. In: Ghahramani Z, Welling M, Cortes C, Lawrence ND, Weinberger KQ (eds) *Advances in neural information processing systems*, 2014, vol 27. Curran Associates, Inc., Red Hook, pp 2672–2680. <http://papers.nips.cc/paper/5423-generative-adversarial-nets.pdf>
- Hajdik HM, Yildirim A, Wu E, Brelje BJ, Seraj S, Mangano M, Anibal JL, Jonsson E, Adler EJ, Mader CA, Kenway GKW, Martins JRRA (2023) pyGeo: a geometry package for multidisciplinary design optimization. *J Open Source Softw* 8(87):5319. <https://doi.org/10.21105/joss.05319>
- Han Z, Xu C, Zhang L, Zhang Y, Zhang K, Song W (2019) Efficient aerodynamic shape optimization using variable-fidelity surrogate models and multilevel computational grids. *Chin J Aeronaut*. <https://doi.org/10.1016/j.cja.2019.05.001>
- He P, Mader CA, Martins JRRA, Maki KJ (2018) An aerodynamic design optimization framework using a discrete adjoint approach with OpenFOAM. *Comput Fluids* 168:285–303. <https://doi.org/10.1016/j.compfluid.2018.04.012>
- He X, Li J, Mader CA, Yildirim A, Martins JRRA (2019) Robust aerodynamic shape optimization—from a circle to an airfoil. *Aerosp Sci Technol* 87:48–61. <https://doi.org/10.1016/j.ast.2019.01.051>
- Jacome LB, Elham A (2017) Wing aerostructural optimization under uncertain aircraft range and payload weight. *J Aircr* 54(3):1109–1120. <https://doi.org/10.2514/1.c034050>
- Jameson A (1988) Aerodynamic design via control theory. *J Sci Comput* 3(3):233–260. <https://doi.org/10.1007/bf01061285>
- Kenway GKW, Martins JRRA (2014) Multipoint high-fidelity aerostructural optimization of a transport aircraft configuration. *J Aircr* 51(1):144–160. <https://doi.org/10.2514/1.C032150>

- Kenway GKW, Martins JRRA (2016) Multipoint aerodynamic shape optimization investigations of the Common Research Model wing. *AIAA J* 54(1):113–128. <https://doi.org/10.2514/1.J054154>
- Kenway GKW, Mader CA, He P, Martins JRRA (2019) Effective adjoint approaches for computational fluid dynamics. *Prog Aerosp Sci*. <https://doi.org/10.1016/j.paerosci.2019.05.002>
- Koziel S, Leifsson L (2013) Surrogate-based aerodynamic shape optimization by variable-resolution models. *AIAA J* 51(1):94–106. <https://doi.org/10.2514/1.j051583>
- LeCun Y, Bengio Y, Hinton G (2015) Deep learning. *Nature* 521(7553):436–444. <https://doi.org/10.1038/nature14539>
- Li J, Zhang M (2021) Data-based approach for wing shape design optimization. *Aerosp Sci Technol* 112:106639. <https://doi.org/10.1016/j.ast.2021.106639>
- Li J, Zhang M (2021) On deep-learning-based geometric filtering in aerodynamic shape optimization. *Aerosp Sci Technol* 112:106603. <https://doi.org/10.1016/j.ast.2021.106603>
- Li J, Zhang M (2021) Adjoint-free aerodynamic shape optimization of the common research model wing. *AIAA J* 59(6):1990–2000. <https://doi.org/10.2514/1.j059921>
- Li J, Bouhlel MA, Martins JRRA (2019) Data-based approach for fast airfoil analysis and optimization. *AIAA J* 57(2):581–596. <https://doi.org/10.2514/1.j057129>
- Li J, He S, Martins JRRA (2019) Data-driven constraint approach to ensure low-speed performance in transonic aerodynamic shape optimization. *Aerosp Sci Technol* 92:536–550. <https://doi.org/10.1016/j.ast.2019.06.008>
- Li J, Zhang M, Martins JRRA, Shu C (2020) Efficient aerodynamic shape optimization with deep-learning-based geometric filtering. *AIAA J* 58(10):4243–4259. <https://doi.org/10.2514/1.j059254>
- Li J, Du X, Martins JRRA (2022) Machine learning in aerodynamic shape optimization. *Prog Aerosp Sci* 134:100849. <https://doi.org/10.1016/j.paerosci.2022.100849>
- Li J, He S, Zhang M, Martins JRRA, Khoo BC (2022) Physics-based data-driven buffet-onset constraint for aerodynamic shape optimization. *AIAA J* 60(8):4775–4788. <https://doi.org/10.2514/1.J061519>
- Li J, Zhang M, Tay CMJ, Liu N, Cui Y, Chew SC, Khoo BC (2022) Low-Reynolds-number airfoil design optimization using deep-learning-based tailored airfoil modes. *Aerosp Sci Technol* 121:107309. <https://doi.org/10.1016/j.ast.2021.107309>
- Li J, Dao MH, Le QT (2024) Data-driven modal parameterization for robust aerodynamic shape optimization of wind turbine blades. *Renew Energy* 224:120115. <https://doi.org/10.1016/j.renene.2024.120115>
- Liem RP, Kenway GKW, Martins JRRA (2015) Multimission aircraft fuel burn minimization via multipoint aerostructural optimization. *AIAA J* 53(1):104–122. <https://doi.org/10.2514/1.J052940>
- Lyu Z, Xu Z, Martins JRRA (2014) Benchmarking optimization algorithms for wing aerodynamic design optimization. In: *Proceedings of the 8th international conference on computational fluid dynamics*, Chengdu, Sichuan, China, 2014. ICCFD8-2014-0203
- Lyu Z, Kenway GKW, Martins JRRA (2015) Aerodynamic shape optimization investigations of the Common Research Model wing benchmark. *AIAA J* 53(4):968–985. <https://doi.org/10.2514/1.J053318>
- Lyu Y, Seth A, Liem RP (2024) Fuel consumption and trailing-edge noise tradeoff studies via mission-based airfoil shape optimization. *Aerosp Sci Technol* 151:109331. <https://doi.org/10.1016/j.ast.2024.109331>
- Mader CA, Martins JRRA, Alonso JJ, Weide E (2008) ADjoint: an approach for the rapid development of discrete adjoint solvers. *AIAA J* 46(4):863–873. <https://doi.org/10.2514/1.29123>
- Mader CA, Kenway GKW, Yildirim A, Martins JRRA (2020) ADflow: an open-source computational fluid dynamics solver for aerodynamic and multidisciplinary optimization. *J Aerosp Inf Syst* 17(9):508–527. <https://doi.org/10.2514/1.j010796>
- Madsen MHA, Zahle F, Sørensen NN, Martins JRRA (2019) Multipoint high-fidelity CFD-based aerodynamic shape optimization of a 10 MW wind turbine. *Wind Energy Sci* 4:163–192. <https://doi.org/10.5194/wes-4-163-2019>
- Martins JRRA (2022) Aerodynamic design optimization: challenges and perspectives. *Comput Fluids* 239:105391. <https://doi.org/10.1016/j.compfluid.2022.105391>
- Nocedal J, Wright SJ (2006) *Numerical optimization*. Springer, New York, pp 448–579. <https://doi.org/10.1007/978-0-387-40065-5>
- Pang X, Cao S (2024) Optimization of fin arrangement in solar thermal storage devices and convolutional neural network modeling. *Int Commun Heat Mass Transf* 156:107670. <https://doi.org/10.1016/j.icheatmasstransfer.2024.107670>
- Queipo NV, Haftka RT, Shyy W, Goel T, Vaidyanathan R, Kevin Tucker P (2005) Surrogate-based analysis and optimization. *Prog Aerosp Sci* 41(1):1–28. <https://doi.org/10.1016/j.paerosci.2005.02.001>
- Reuther JJ, Jameson A, Alonso JJ, Rimlinger MJ, Saunders D (1999) Constrained multipoint aerodynamic shape optimization using an adjoint formulation and parallel computers, Part 1. *J Aircr* 36(1):51–60. <https://doi.org/10.2514/2.2413>
- Wang Y, Shimada K, Barati Farmani A (2023) Airfoil GAN: encoding and synthesizing airfoils for aerodynamic shape optimization. *J Comput Des Eng* 10(4):1350–1362. <https://doi.org/10.1093/jcde/qwad046>
- Wei Z, Yang A, Li J, Bauerheim M, Liem RP, Fua P (2024) DeepGeo: deep geometric mapping for automated and effective parameterization in aerodynamic shape optimization. In: *AIAA AVIATION FORUM AND ASCEND 2024*, 2024. American Institute of Aeronautics and Astronautics, Las Vegas. <https://doi.org/10.2514/6.2024-3839>
- Wu E, Kenway G, Mader CA, Jasa J, Martins JRRA (2020) pyOptSparse: a Python framework for large-scale constrained nonlinear optimization of sparse systems. *J Open Source Softw* 5(54):2564. <https://doi.org/10.21105/joss.02564>
- Yildirim A, Kenway GKW, Mader CA, Martins JRRA (2019) A Jacobian-free approximate Newton–Krylov startup strategy for RANS simulations. *J Comput Phys* 397:108741. <https://doi.org/10.1016/j.jcp.2019.06.018>
- Yu Y, Lyu Z, Xu Z, Martins JRRA (2018) On the influence of optimization algorithm and initial design on wing aerodynamic shape optimization. *Aerosp Sci Technol* 75:183–199. <https://doi.org/10.1016/j.ast.2018.01.016>
- Zhang Y, Sung WJ, Mavris DN (2018) Application of convolutional neural network to predict airfoil lift coefficient. In: *2018 AIAA/ASCE/AHS/ASC structures, structural dynamics, and materials conference*, 2018. American Institute of Aeronautics and Astronautics, Reston. <https://doi.org/10.2514/6.2018-1903>

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